

Hypothesis of 3-Brane-Higher-Dimensional Resonance Coupling

A unified thought-experiment model for ghosts, paranormal sightings, past-life memories and the illusion of reincarnation based on brane-universe ideas

Document Note: This text is a theoretical deduction for thought experiments. It draws on the conceptual framework of string-theory brane cosmology. **It is not a mainstream physics theory verified by experiments.** It is used solely for logical reasoning to interpret phenomena such as ghost sightings and alleged past-life memories.

I. Core Axioms of the Model (Fundamental Assumptions, including newly-added features: energy-dependent resonance strength, projection distance attenuation, coexistence of multiple resonant configurations, and species-specific resonance selectivity)

1. Our world is a 3-dimensional brane universe embedded in a higher-dimensional bulk space. Ordinary matter, electromagnetic waves and photons are confined inside the 3-brane and cannot enter higher-dimensional space directly.
2. **Resonance strength determined by energy density:** The higher the energy density of biological neural activity, the stronger the amplitude of the excited higher-dimensional resonant field. Energy consumption varies greatly across human tissues: muscles and bones consume little energy and can hardly excite effective higher-dimensional resonance. The brain is the human organ with the highest energy density. Its neural electrical activity continuously excites coherent resonant configurations in the higher-dimensional bulk space. During life, consciousness, memory and mental states exist simultaneously in both the neural activity within the 3-brane brain and the higher-dimensional resonant configurations; the two form an integrated whole.
3. The intensity of human fixation and subjective belief regulates the coherent stability of higher-dimensional resonant fields. Stronger fixation locks the phase of resonant waveforms more firmly and enhances anti-disturbance capacity; weaker fixation results in looser resonant binding.
4. After biological death, neural electrical activity in the 3-brane brain ceases and no longer supplies the driving source for resonance. Nevertheless, already-formed higher-dimensional coherent resonant configurations do not vanish immediately. External energy fields within the 3-brane act as environmental disturbance sources. Together with higher-dimensional

background fluctuations, they introduce entropy-increase effects, gradually disordering and dispersing resonant waveforms.

5. **Influence of environmental energy density on resonance preservation**

- The higher the environmental energy density, the stronger the disturbance exerted on higher-dimensional resonant configurations; resonance is more readily damaged, and decoherence proceeds faster.
- In industrial cities, power-generation plants and large-scale power-facility zones, high electromagnetic and energy densities strongly perturb higher-dimensional resonant waveforms. Resonant configurations left by the deceased can hardly be preserved over long timescales.
- In remote, sparsely-populated regions such as deep mountains and high-altitude plateaus with minimal anthropogenic industrial-energy input, background disturbances are weak. Higher-dimensional resonant configurations can persist longer, raising the probability of trans-brane resonant coupling.
- Extreme high-energy events such as nuclear-weapon detonations generate intense energy oscillations that violently impact higher-dimensional resonant fields and can instantly destroy established higher-dimensional resonant configurations by forcing them into decoherence.

6. **Higher-dimensional resonance in animals and species-specific frequency**

isolation: Animals can also excite higher-dimensional resonance via brain-neural activity. However, animal brains are smaller, their neural-circuit complexity is lower, and the energy density and coherence of their neural activity are weaker. Consequently, they produce low-amplitude, poorly-stable resonance. After animal death, corresponding higher-dimensional resonant configurations decohere extremely rapidly and seldom persist. Different species possess exclusive resonant-frequency bands, analogous to language barriers: animal brains respond preferentially to conspecific resonance, while human brains are most sensitive to human-specific resonant frequencies. Humans barely perceive animal resonance and animals barely perceive human resonance, much as humans cannot understand animal vocal communication.

7. **Distance-attenuation principle: projection effect of higher-dimensional**

fields onto the 3-brane: The influence exerted by higher-dimensional resonant

configurations within our 3-dimensional world manifests fundamentally as their projection onto the 3-brane. Geographically close to the resonant configuration, projections are sharp and signals are strong; with increasing distance, projections become distorted and blurred, and signals decay accordingly. Generally speaking, trans-brane resonance coupling occurs more readily at short geographic distances. Exceptions exist: certain individuals possess intrinsically higher resonance sensitivity and may detect faint projected resonance even at considerable distances.

8. **Rule for co-existing multiple resonant configurations:** Higher-dimensional space can accommodate massive volumes of low-dimensional information and can in principle store data for all resonant configurations simultaneously. Resonant configurations produce no superimposed interference among themselves. Multiple resonant groups can coexist in higher-dimensional space without disrupting one another's phases. Disturbances originate solely from external-energy perturbations penetrating from the 3-brane into higher dimensions, for example industrial electromagnetic radiation or high-energy shocks from nuclear detonations. High-energy events do not arise from interference between resonances; instead, external energy directly impacts and destroys the coherent structure of higher-dimensional resonance.
9. **Decay rule:** Stronger pre-mortem fixation enhances a higher-dimensional resonant configuration's resistance to decoherence and extends its lifetime. With weak fixation, resonant structures rapidly descend into chaotic dissipation.
10. **Trans-brane resonance-interaction rule:** Trans-brane coupling channels open only when the oscillating frequency of a living human brain falls within the same resonant window as the intrinsic frequency of a higher-dimensional resonant configuration. Higher-dimensional fields do not emit visible-light photons within the 3-brane. Once coupling is established, higher-dimensional fields directly modulate neural electrical signals inside the human brain and generate subjective perceptions such as hallucinated visions and sounds within consciousness, without requiring light to enter the eyes. Besides environmental conditions, successful reception of resonance also depends on the recipient's individual neural constitution and electroencephalographic characteristics.
11. **Reincarnation as an illusion:** If incompletely dissipated higher-dimensional resonant configurations approach newborn infants and match the oscillating brain-wave frequencies of the developing infant brain, trans-brane resonance copies memory fragments and emotional segments into the highly-plastic infant

neural circuits. The original higher-dimensional resonant configuration continues its slow entropy-driven dissipation in higher-dimensional space; no complete soul transfers.

Inference: Trans-brane information copying is more likely in high-plateau and remote regions with low industrial-energy input. This corresponds to folk accounts of tul-ku-style alleged reincarnation and divinely-bestowed epics. For instance, Tibetan Gesar epic bards' reputed "divinely-inspired chanting" may be interpreted as follows: in an environment with low background disturbance, ancient higher-dimensional resonant configurations undergo trans-brane resonance with the bard's brain, coupling historical fragments and epic information fragments into the bard's neural field.

II. Full Evolutionary Stages of the Model

Stage 1: Living Organism

Human brains continuously generate neural electrical activity. Supported by high brain energy density, they sustain coherent resonant waveforms in higher-dimensional bulk space. Consciousness and memory reside simultaneously within the 3-dimensional physical brain and higher-dimensional resonant structures. Strong fixation reinforces resonant phase locking and improves structural stability. Tissues such as muscles and bones consume little energy and can barely sustain viable higher-dimensional resonance. Since higher-dimensional resonant fields exist outside the 3-brane, they generate no electromagnetic radiation within 3-dimensional space and cannot be directly observed by ordinary optical or electromagnetic instruments.

Stage 2: Physical Death and Formation of Higher-Dimensional Resonant Apparitional Configurations

Biological brain electrical activity terminates, removing the driving source sustaining resonance. Nevertheless, coherent resonant waveforms in higher-dimensional space retain pre-established phase locking and maintain ordered configurations for a finite time; these are defined as "apparitions" within this model. Such higher-dimensional resonant configurations preserve memory fragments, emotional traits and intrinsic oscillating frequencies from the deceased. Coherence is highest immediately after death, raising the probability of trans-brane resonant coupling with living humans whose brain-wave frequencies match. Environmental energy density directly governs configuration lifetime: high-energy urban settings erode resonance rapidly, whereas low-disturbance mountain-plateau environments greatly prolong persistence. Higher-dimensional resonance projects onto the 3-brane: projections are sharp nearby and decay into blurriness at distance. Sensitive individuals may detect faint projected signals from remote locations.

Stage 3: Entropy-Driven Decoherence and Dissipation of Higher-Dimensional Resonant Configurations

Deprived of continuously supplied ordered oscillation from the brain, 3-dimensional environmental-energy disturbances together with higher-dimensional background noise and spacetime perturbations persistently break phase synchronisation of resonant waveforms (entropy-increase principle).

1. **Early phase:** Coherence declines gradually; trans-brane coupling strength weakens, and memories and personality information grow progressively blurred. Each resonant interaction with living humans introduces additional disturbance and accelerates dissipation of resonant configurations.
2. **Late phase:** Waveform phases become fully disordered; coherent structures disintegrate, unified resonant frequencies vanish, and trans-brane coupling with human brains within the 3-brane becomes impossible. The configuration becomes completely undetectable.

Extreme high-energy shocks such as nuclear detonations bypass gradual entropy-driven decay and shatter resonant coherent structures instantaneously.

Note: Higher-dimensional fields themselves do not disappear. Only ordered oscillatory structures encoding personality and memory are destroyed and dissolve into higher-dimensional background noise. Numerous resonant configurations may coexist in higher-dimensional space without mutual superimposed interference. Only external high-energy perturbations originating from the 3-dimensional world can damage resonance.

III. Explanations for Observed Phenomena under the Model

1. **Apparition sightings:** Higher-dimensional resonant configurations project onto the 3-brane. Upon trans-brane resonant coupling with living humans whose brain-wave frequencies match, they directly modulate the visual and auditory cortex and generate hallucinated human-shaped figures, whispers and other subjective percepts.
 - Undetectable by ordinary cameras: No visible-light photons are emitted within the 3-brane, so optical equipment cannot record the phenomenon.
 - Individual differences: Only persons whose brain-waves lie within the resonant window experience sightings, while other onlookers perceive nothing. Perceptual outcomes depend on the recipient's neural

constitution.

- Distance effect: Projections are sharper and detection probability higher at short geographic distances; a small subset of highly-sensitive individuals may detect faint projections remotely.
 - Geographic variation: In remote mountain and high-plateau regions with low anthropogenic-energy interference, resonant configurations persist longer and sightings are more probable. Observations are rare in cities and near power plants where energy-density levels are high.
 - More frequent sightings of the recently deceased: Resonant coherence peaks shortly after death, maximising coupling probability.
2. **Fixation effects on persistence duration:** Strong pre-mortem fixation and emotion reinforce phase locking of higher-dimensional resonant waveforms and slow entropy-driven decoherence. When fixation is weak, post-mortem resonant configurations disintegrate rapidly and are seldom observable.
 3. **Weak animal resonance and species-specific frequency isolation:** Animal brains are small with low neural complexity. Insufficient energy density in neural activity produces faint, unstable higher-dimensional resonance that dissipates quickly after animal death. Trans-brane coupling detectable by humans rarely occurs. Human and animal resonances occupy distinct frequency bands analogous to linguistic barriers: human brains prioritise human-origin resonance, animal brains prioritise conspecific resonance, and cross-species perception is difficult.
 4. **Alleged past-life memories, tul-ku-style reincarnation phenomena and divinely-bestowed epics:** In high-altitude plateaus such as the Tibetan-Qinghai Plateau with minimal anthropogenic industrial-energy interference, environmental perturbations are weak and ancient higher-dimensional resonant configurations can survive for long periods. When the brain-wave frequency of a nearby developing newborn matches, trans-brane resonance copies memory fragments and emotional states.
 - Only information copies are produced; no ontological soul transfer occurs.
 - Memories are fragmentary and tend to fade during childhood as higher-dimensional resonant configurations dissipate via entropy increase.

- Gesar epic bards' "divinely-inspired chanting": Ancient epic-related higher-dimensional resonant configurations persist in low-disturbance plateau environments. When the bard enters a special brain-wave state, trans-brane resonance is triggered and epic-content fragments are received.
 - In principle, a single higher-dimensional resonant configuration may couple sequentially with multiple infants or sensitive individuals, producing similar memory fragments across different subjects.
5. **Locations prone to paranormal phenomena:** Local gravitational perturbations, dark-energy background fluctuations within the 3-brane, together with low anthropogenic-energy disturbance, reduce background noise for higher-dimensional fields and slow decoherence. Higher-dimensional resonant configurations survive longer, giving rise to persistently "haunted" locations.
 6. **Destructive effects of extreme high-energy events:** Large-scale intense-energy oscillations generated by nuclear detonations penetrate the 3-brane and impact higher-dimensional space, destroying all local higher-dimensional resonant configurations instantaneously. Destruction arises not from resonance-versus-resonance interference but from external high-energy perturbations originating in 3-dimensional spacetime.
 7. **Coexistence of multiple apparitional-like configurations:** Higher-dimensional space can host countless resonant configurations simultaneously. Resonances produce no mutual superimposed interference or mutual destruction. Provided no external high-energy disturbance occurs, large numbers of resonant configurations may coexist within the same higher-dimensional locality.

IV. Thought-Experiment Detection Scheme for Resonance (The verification plan is still not mature)

Based on this hypothesis, a thought-experiment setup can be devised to detect trans-brane higher-dimensional resonant fields:

1. Shielded chamber: Faraday cage combined with vibration-acoustic insulation to isolate all known electromagnetic and vibrational interference originating in the 3-dimensional world.
2. Tunable resonant probe: Continuous wide-range frequency sweeping to search for narrow-band resonant signals produced by trans-brane coupling.
3. Simultaneous electroencephalogram (EEG) recording for human volunteers, to

distinguish two scenarios:

- Resonant signal detected by the probe + visual-cortex evoked potentials on the volunteer's EEG together with subjective reports of hallucinations → tentative evidence for trans-brane higher-dimensional resonant coupling.
- Subjective hallucinations occur with no corresponding resonant signal from the probe → purely endogenous brain hallucination unrelated to higher-dimensional resonant fields.

4. Controlled-experiment groups:

- **Group A:** Run identical tests in high-electromagnetic-interference urban environments and low anthropogenic-energy mountain locations; compare resonant-signal-detection probabilities.
- **Group B:** Use tDCS transcranial direct-current stimulation to artificially modulate subjects' brain-wave frequencies and toggle resonant windows. Test whether the capacity to perceive higher-dimensional resonant information varies with brain-wave frequency.
- **Group C:** Distance-gradient comparison experiment. Test how distance affects projection clarity and perceived signal strength, while screening for sensitive subjects to verify remote-detection feasibility.
- **Group D:** Cross-species comparison experiment. Compare human and animal responses to species-specific resonant signals to test the frequency-isolation hypothesis.

V. Model Premises and Intrinsic Limitations

Additional Required Assumptions (Not part of established standard physics)

1. The energy density of human-brain neural activity determines excitation strength for higher-dimensional resonant fields; high brain-energy output is a prerequisite for viable higher-dimensional resonance.
2. Weak neural electromagnetic oscillations of the human brain can excite stable coherent resonant configurations in higher-dimensional bulk space.
3. Electromagnetic and energy fields within the 3-brane can penetrate brane boundaries and perturb higher-dimensional resonant waveforms; extreme high-energy events are capable of destroying coherent higher-dimensional resonant configurations.

4. Higher-dimensional resonant fields project onto the 3-brane and exhibit distance-dependent attenuation: projections grow more blurred at greater distances. Certain sensitive individuals can receive faint remote projections.
5. Higher-dimensional space permits coexistence of large numbers of resonant configurations with no intrinsic superimposed interference among them. Perturbations stem only from external energy originating in the 3-brane.
6. Species-specific resonant-frequency isolation exists: human brains prioritise human-origin resonant frequencies and animal brains prioritise conspecific resonant frequencies.
7. Higher-dimensional resonant fields can encode and store complex memory, personality and emotional information over finite timescales.
8. Trans-brane coupling exhibits selectivity, and reception quality correlates with the recipient's individual neural constitution.
9. The second law of thermodynamics (entropy increase) holds in higher-dimensional bulk space; resonant waveforms undergo decoherence over time.

Unresolved Gaps within the Model

1. Quantitative projection-attenuation relationship: The mathematical law governing how higher-dimensional projections decay with geographic distance remains undetermined.
2. Energy-coupling channel: The underlying physical mechanism whereby 3-dimensional-world energy penetrates the brane and acts upon higher-dimensional fields requires further elaboration.
3. Physiological basis for sensitive individuals: The special neural properties of persons capable of remote resonance perception await explanation.
4. Boundaries for species-specific resonant bands: Criteria defining frequency boundaries between different-species resonances, and which animal-brain energy-density thresholds yield preservable higher-dimensional resonance, remain undefined.

VI. Conclusion

This hypothesis integrates apparition-related phenomena, paranormal sightings, alleged past-life memories and plateau-region divinely-best-epic accounts within one self-consistent framework. An apparition is essentially a coherent higher-dimensional

resonant configuration left behind after human biological death. Resonance excitation strength is governed by the energy density of biological brain-neural activity. Muscles and bones produce negligible resonance. Animal brains possess insufficient energy density, generating weak, rapidly-dissipating resonance, with frequency-barrier separation between human and animal resonance. Higher-dimensional resonance manifests in the 3-dimensional world as a projection effect: projections are sharper at short range and decay with increasing distance, and a minority of sensitive individuals can detect signals remotely. Countless resonant configurations can coexist in higher-dimensional space with no mutual interference. Only external-energy perturbations originating from our 3-dimensional world can damage resonant structures. High-energy-density environments such as cities and power plants continuously perturb higher-dimensional resonance and accelerate dissipation. In remote mountain and high-plateau regions with minimal anthropogenic-energy interference, higher-dimensional resonant configurations survive longer and trans-brane resonant-interaction events are more probable. Extreme high-energy explosions can instantly destroy all local higher-dimensional resonant configurations. Fixation strength determines anti-disturbance robustness of resonant configurations. Trans-brane resonant coupling can directly modulate human neural electrical activity to produce hallucinations, and can copy memory fragments into newborn infants, generating illusory phenomena interpreted as reincarnation or divinely-revealed information. The whole model obeys the speed-of-light limit, causality and light-cone rules valid inside the 3-brane universe. Information exchange across dimensions occurs exclusively when resonant-matching conditions are satisfied.